A CARTESIAN SQUARE WITH AN EPIMORPHISM, OR A COCARTESIAN SQUARE WITH A MONOMORPHISM

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file homology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument for the first assertion, with each cited lemma applied to a specific displayed sequence; the closing clause 'and (2) follows by duality' is the author's own routine duality remark on the part written out in full inside the counted span, and it is preserved verbatim.

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Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the comment-type reference, slogan and history environments (native preamble lines 8-13), the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217, the actual last line of the file).

Allowed standard prerequisites (editorial).

- Characterization of cartesian and cocartesian squares by exactness. Same chapter, Lemma lemma-characterize-cartesian. Supplies the translation of cartesianness and cocartesianness of the square into exactness of the displayed sequence through the direct sum, applied in both directions in the proof.
- Kernels and cokernels in a cartesian square. Same chapter, Lemma lemma-cartesian-kernel. Supplies that the morphism induced on the kernels of the two horizontal maps is an isomorphism, which produces the epimorphism in the last step.
- Characterizations of monomorphisms and epimorphisms. Same chapter, Lemma lemma-characterize-injective. Converts the isomorphism obtained from the previous lemma into the statement that the relevant morphism is an epimorphism.

Source reference. Original lemma 08N2: sources/stacks-homology/pinned-original-source.tex, source lines 1026–1026. Navigation pointer invoked at proof line 1127; the counted span contains the pointer only and the cited result is not reproduced

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Source reference. Original lemma 08N3: sources/stacks-homology/pinned-original-source.tex, source lines 1069–1069. Navigation pointer invoked at proof line 1130; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 010C: sources/stacks-homology/pinned-original-source.tex, source lines 870–870. Navigation pointer invoked at proof line 1131; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let $\mathcal{A}$ be an abelian category. Let*

$$\begin{array}{ccc} w & \xrightarrow{f} & y \\ {\scriptstyle g}\downarrow & & \downarrow{\scriptstyle h} \\ x & \xrightarrow{k} & z \end{array}$$

be a commutative diagram.

(1) *If the diagram is cartesian and k is an epimorphism, then the diagram is cocartesian and f is an epimorphism.*
(2) *If the diagram is cocartesian and g is a monomorphism, then the diagram is cartesian and h is a monomorphism.*

Proof. Suppose the diagram is cartesian and k is an epimorphism. Let $u = (g, f) : w \to x \oplus y$ and let $v = (k, -h) : x \oplus y \to z$. As k is an epimorphism, v is an epimorphism, too. Therefore and by Lemma 08N2, the sequence $0 \to w \xrightarrow{u} x \oplus y \xrightarrow{v} z \to 0$ is exact. Thus, the diagram is cocartesian by Lemma 08N2. Finally, f is an epimorphism by Lemma 08N3 and Lemma 010C. This proves (1), and (2) follows by duality. □